



Project ID:
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01 THE PROBLEM



- ⚠️ Phishing attacks imitate trusted brands and use urgent wording.
- ⚠️ Attackers hide malicious intent behind misleading URLs and fake login or payment requests.
- ⚠️ Mobile users find suspicious links harder to inspect.
- ⚠️ Blacklists alone are weak against newly registered or zero-day phishing domains.

02 PROJECT GOAL

- ✓ Combine AI classification with explainable, identity-based evidence.
- ✓ Work with suspicious messages and standalone URLs.
- ✓ Be designed for non-technical mobile users.
- ✓ Return a clear label, risk score, and short explanation.
- ✓ Help users make safer decisions before opening a link.

03 SCREENS

MsgGuard

Message Sharing and Analysis

Paste a message or URL, or share content from another app for phishing analysis.

Suspicious message or URL

Urgent: Your PayPal account has been temporarily locked due to suspicious activity. Verify your identity now to avoid permanent suspension: <https://deepesek.com/paypal-security/verify-account>

Analyze

How it works

- ✓ Submit message or URL.
- ✓ MsgGuard checks content, URL, and website identity.
- ✓ Identifies risks and final explanation

Risk Overview

High Risk

80%

Top Reasons

- Domain generic updates meaning urgency wording present.
- Detected spoofed brand: DHL-like, inconsistent dns and suspicious URL pattern
- URL short path suspiciously long with multiple redirects
- Identity similarity compromised using similar-looking brand name

View Details >

Close

04 KEY TECHNOLOGIES

	Frontend	React Native, Expo
	Backend	Flask, Python
	AI Model	Fine-tuned BERT
	Domain Checks	WHOIS, IP Geolocation, Brand-Domain Matching
	Development	VS Code, MongoDB

05 RESULTS



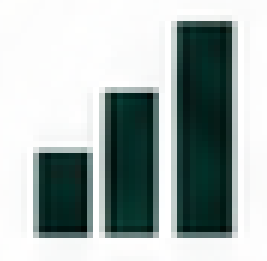
99.70%
Accuracy



99.40%
Precision



100.00%
Recall



99.70%
F1-score



0.15
seconds
Average Response Time



1000
Total Valid Samples



Evaluated on 1,000 external test cases (500 legitimate, 500 phishing) including real-world messages, malicious URLs, legitimate messages and edge cases.

06 CHALLENGES & SOLUTIONS

Challenge	Solution
⚠️ Combining strong detection with explainability.	✓ Hybrid architecture with BERT plus modular identity-based checks.
⚠️ New domains can be suspicious but also legitimate.	✓ Use domain age, WHOIS, IP location and brand-domain matching as supporting evidence.
⚠️ New phishing variants may differ from training data.	✓ Combine BERT with URL identity checks, brand-domain matching, honeypage detection, and domain evidence to detect new phishing variants beyond the training data.

07 SYSTEM OVERVIEW

